Zhilong Xue

ACADEMIC CURRICULUM VITAE · ANALYSIS AND PARTIAL DIFFERENTIAL EQUATIONS

Academy of Mathematics and Systems Science, Chinese Academy of Sciences · Beijing, China
zhilongxue@amss.ac.cn · www.zhilongxue.cn

RESEARCH PROFILE

Mathematical fluid mechanics: vortex patch dynamics, rotating and time-periodic structures, regularity, and singularity formation in incompressible fluid equations.
Kinetic equations: Landau damping in Vlasov-Poisson systems.

ACADEMIC APPOINTMENT

Jul. 2025 – present **Postdoctoral Researcher**, Academy of Mathematics and Systems Science, Chinese Academy of Sciences
Mentor: Quoc-Hung Nguyen

EDUCATION

Sep. 2020 – Jun. 2025 **PhD in Mathematics**, Beijing Normal University
Supervisor: Liutang Xue

Sep. 2016 – Jun. 2020 **BS in Mathematics**, Chang'an University

PUBLICATIONS AND PREPRINTS

Publications

[1] Taoufik Hmidi, Liutang Xue, and **Zhilong Xue**. *Unified theory on V-states structures for active scalar equations. Advances in Mathematics* **486** (2026), Paper 110750.

[2] Taoufik Hmidi, Liutang Xue, and **Zhilong Xue**. *Doubly connected V-states in geophysical models: a general framework. Calculus of Variations and Partial Differential Equations* **65** (2026), Paper 137.

[3] Changhui Tan, Liutang Xue, and **Zhilong Xue**. *A revisit of patch solutions for the 2D Loglog-Euler type equation.* To appear in *Revista Matemática Iberoamericana*, 2026. arXiv:2510.18759.

[4] Taoufik Hmidi, Liutang Xue, and **Zhilong Xue**. *Emergence of time periodic solutions for the generalized surface quasi-geostrophic equation in the disc. Journal of Functional Analysis* **285** (2023), no. 10, Paper 110142.

Preprints

[5] Haina Li, Quoc-Hung Nguyen and **Zhilong Xue**. *Pointwise wave–damping for linearized Vlasov–Poisson: compactly supported radial equilibria and relativistic applications.* Preprint, 2026. arXiv:2610.26600.

[6] **Zhilong Xue** and Weicheng Zhan. *Uniformly rotating vortex patches with arbitrarily many genuine holes.* Preprint, 2026. arXiv:2608.19627.

[7] Qianyun Miao, Changhui Tan, Liutang Xue, and **Zhilong Xue**. *Local regularity and finite-time singularity for a class of generalized SQG patches on the half-plane.* Preprint, 2024. arXiv:2410.19273.

RESEARCH FUNDING AND HONORS

Research Funding

Jul. 2026 **National Postdoctoral Researcher Funding Program** (Grade B), China
No. GZB20260747

Jul. 2026 **General Program, China Postdoctoral Science Foundation**
No. 2026M793420

Jul. 2023 **Scholarship for Study Abroad**, China Scholarship Council

Honors

Dec. 2022, 2021 **First-class Scholarship**, Beijing Normal University

Dec. 2019 **National Scholarship for Undergraduate Students**, Ministry of Education of China

RESEARCH VISITS

Oct. 2023 – Oct. 2024 **Visiting PhD Student**, University of Seville, Spain
Supervisor: Francisco Gancedo · Funded by the China Scholarship Council

Sep. 2024 – Oct. 2024 **Short-term Research Visit**, University of Granada, Spain
Host: Claudia García

INVITED TALKS & POSTERS

30 Oct. 2026 Invited talk *Uniformly rotating vortex patches with arbitrarily many genuine holes*
Shandong University, Jinan, China

4 Jun. 2026 Invited talk *A Revisit of Patch Solutions for the 2D Loglog-Euler Type Equation*
Singularities, Non-uniqueness and Complex Dynamics of Incompressible Fluids, Bath, United Kingdom

27 May 2026 Poster *A Revisit of Patch Solutions for the 2D Loglog-Euler Type Equation*
Workshop: Analysis of PDEs in Dalian, Dalian, China

16 Oct. 2024 Invited talk *The regularity for generalized SQG patches*
PhD Seminar, IMUS, University of Seville, Seville, Spain

21 Mar. 2024 Invited talk *Unified theory on V-states structures for active scalar equations*
Ecuaciones Diferenciales Seminar, IMAG, University of Granada, Granada, Spain

13 Nov. 2023 Invited talk *Emergence of time periodic solutions for the generalized surface quasi-geostrophic equation in the disc*
Fluid Conversations Seminar, IMUS, University of Seville, Seville, Spain

TEACHING EXPERIENCE

Sep. 2025 – Jan. 2026 **Teaching Assistant**, University of Chinese Academy of Sciences
Course: Linear Algebra.
Led problem-solving sessions and graded homework and exams.

Sep. 2020 – Jun. 2023 **Teaching Assistant**, Beijing Normal University
Courses: Calculus I and II, Complex Analysis, and Ordinary Differential Equations.
Led problem-solving sessions, answered students' questions, and graded homework and exams.

SELECTED CONFERENCES

Aug. 2026 *Advances in Mathematical Fluid Dynamics 2026*, Peking University, China

Aug. 2026 *Workshop on Nonlinear PDEs in Fluid Mechanics*, Chinese Academy of Sciences, China

Jul. 2025 *Mathematical Theory on Navier-Stokes Equation*, Peking University, China

Jun. 2025 *Advances in Mathematical Fluid Dynamics*, Peking University, China

Nov. 2024 *Mixing, Enhanced Dissipation and Stability Effects in Fluid Dynamics*, Kunming, China

Jul. 2024 *Alhambra PDE Days*, IMAG, University of Granada, Spain

Jul. 2024 *9th European Congress of Mathematics*, University of Seville, Spain